

**cardona-qft-methods — formula report**

3760 inline formulas (section omitted; all rendered), 1012 display equations, 8 tables, 40 unrecovered image regions.

Unresolved (1)

| Identifier                                                                                                                                                                                                                                                                                                           | Page | Conf. | LaTeX source                                                                                                                                                                                                                                                                                                                        | Rendered       | Scan image |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|
| cardona-qft-methods_<br>E08654                                                                                                                                                                                                                                                                                       | 261  | —     | $\phi_{\theta}(x)\phi_{\theta}(y)=e^{-\frac{1}{2}\frac{\partial}{\partial x^{\mu}}\frac{\partial}{\partial y^{\nu}}\theta^{\mu\nu}}\phi_0(x)\phi_0(y)e^{\frac{1}{2}\left(\frac{\overleftarrow{\partial}}{\partial x^{\mu}}+\frac{\overrightarrow{\partial}}{\partial y^{\mu}}\right)\theta^{\mu\nu}P_{\nu}}\neq\phi_0(x)\phi_0(y).$ | (not rendered) |            |
| <b>Conf.</b> MathPix confidence: <span>■</span> $\geq 0.9$ <span>■</span> 0.5-0.9 <span>■</span> $< 0.5$ — no value<br><b>Residual</b> render vs scan (inkdrill): <span>●</span> C component <span>●</span> W weak <span>●</span> S stable <span>●</span> N noise <span>●</span> K clean <span>●</span> not measured |      |       |                                                                                                                                                                                                                                                                                                                                     |                |            |

**Flagged, not acted on — 2 of 90 shown**

| Identifier                                                                                                                                                                             | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                     | Rendered                                                                                                                                                                    | Scan image                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cardona-qft-methods_<br>EQ0077 • C +44                                                                                                                                                 | 046  | ■0.150 | <pre>\left.\begin{array}{c} 1 &amp; \rightarrow &amp; \\ \operatorname{Inn}(\mathcal{A}) &amp; \rightarrow &amp; \\ \operatorname{Aut}(\mathcal{A}) &amp; \rightarrow &amp; \\ \operatorname{Aut}(\mathcal{A}) / \operatorname{Inn}(\mathcal{A}) &amp; \rightarrow &amp; \end{array} \right\} \quad \begin{array}{c} 1, \\ \mathcal{G} \\ \mathcal{G} \rtimes \operatorname{Diff}(M) \\ \operatorname{Diff}(M) \end{array}</pre> | (not rendered)                                                                                                                                                              | $\begin{array}{c} 1 \rightarrow \operatorname{Inn}(\mathcal{A}) \rightarrow \operatorname{Aut}(\mathcal{A}) \rightarrow \operatorname{Aut}(\mathcal{A})/\operatorname{Inn}(\mathcal{A}) \rightarrow 1, \\ 1 \rightarrow \mathcal{G} \rightarrow \mathcal{G} \rtimes \operatorname{Diff}(M) \rightarrow \operatorname{Diff}(M) \rightarrow 1. \end{array}$ |
| cardona-qft-methods_<br>EQ0748 • C -27                                                                                                                                                 | 284  | ■0.547 | <pre>\begin{aligned} A \, d \, S_5 &amp; \, \&amp; \, : \, \&amp; \, -R^2 \, \&amp; \\ = -Z_0^2 + Z_1^2 + Z_2^2 + Z_3^2 + Z_4^2 - Z_5^2 &amp; \, \&amp; \, S^5 : \, \&amp; \, R^2 = Y_1^2 + Y_2^2 + Y_3^2 + Y_4^2 + Y_5^2 + Y_6^2 \\ \&amp; \, S^5 \, \&amp; \, : \, \&amp; \, R^2 &amp; = Y_1^2 + Y_2^2 + Y_3^2 + Y_4^2 + Y_5^2 + Y_6^2 \end{aligned}</pre>                                                                     | $\begin{array}{l} AdS_5 : \quad -R^2 = -Z_0^2 + Z_1^2 + Z_2^2 + Z_3^2 + Z_4^2 - Z_5^2 \\ S^5 : \quad \quad R^2 = Y_1^2 + Y_2^2 + Y_3^2 + Y_4^2 + Y_5^2 + Y_6^2 \end{array}$ | $\begin{array}{l} AdS_5 : \quad \quad -R^2 = -Z_0^2 + Z_1^2 + Z_2^2 + Z_3^2 + Z_4^2 - Z_5^2, \\ S^5 : \quad \quad \quad R^2 = Y_1^2 + Y_2^2 + Y_3^2 + Y_4^2 + Y_5^2 + Y_6^2. \end{array}$                                                                                                                                                                 |
| <b>Conf.</b> MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value<br><b>Residual</b> render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured |      |        |                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                           |

**The remaining 88 flagged rows are stated as a count.**

Each differs from its scan by fewer than 20 components, which is the tail of the distribution rather than its bulk: 18 are component (C), 70 weak (W). By MathPix confidence: 60 at  $\geq 0.9$ , 14 in 0.6–0.9, 14 below 0.6, 0 with none. Out/465 sampled twenty rows drawn from the  $\geq 0.9$  band corpus-wide and found eighteen purely typographic differences, one typographic plus a crop overrun, one borderline and *no* unambiguous content errors — so a high-confidence flag here is evidence of a rendering difference, not of a defect. Every row is in `report.ink.json`.